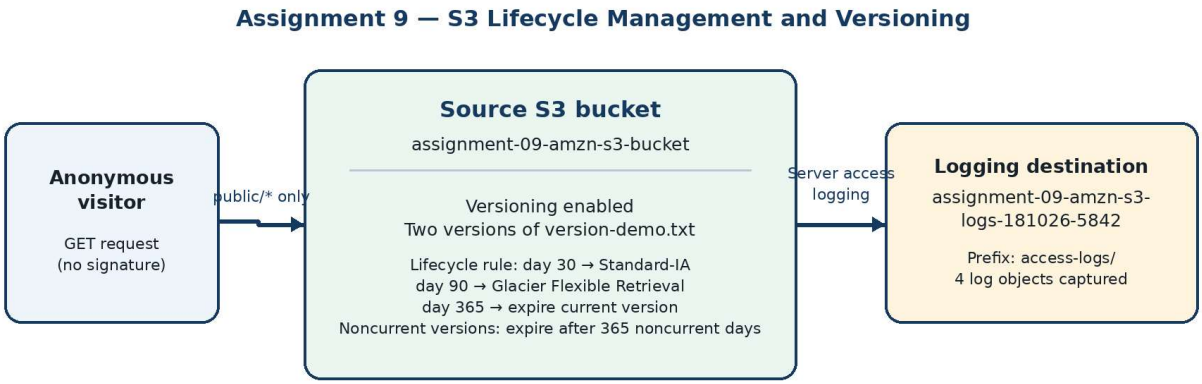
ASSIGNMENT 09

S3 Lifecycle Management and Versioning

AWS & DevOps Practical Report



Buckets were deleted after testing and evidence collection.

Executive Summary

This report documents the implementation and validation of Assignment 9 from the Cloud Engineer Intern Assessment. The exercise used Amazon S3 to create a bucket, enable versioning, configure lifecycle actions, restrict anonymous read access to a designated public/ prefix, and enable server access logging to a separate destination bucket.

The configuration was tested with two versions of a sample object, an anonymous read test of a file under public/, a private-object access test, and successful delivery of four server access log objects. After evidence was saved, server access logging was disabled, the source bucket was emptied and deleted, and the logging destination bucket was emptied and deleted.

1. Objectives and Assessment Requirements

The assessment brief specifies the following tasks and deliverables:

- Create an appropriately named S3 bucket.
- Enable S3 Versioning and demonstrate multiple versions of one object.
- Configure lifecycle transitions to S3 Standard-IA after 30 days and Glacier after 90 days, and expiration after 365 days.
- Configure a bucket policy that allows public read access only for a specific folder/prefix.
- Enable server access logging.
- Submit bucket configuration screenshots, lifecycle policy JSON, a versioning demonstration, and the bucket policy document.

2. Environment and Resource Summary

Item	Configuration
Cloud service	Amazon Simple Storage Service (Amazon S3)
AWS Region	Asia Pacific (Mumbai), ap-south-1
Source bucket	assignment-09-amzn-s3-bucket
Test object for versioning	version-demo.txt (uploaded twice with changed contents)
Public prefix and object	public/ and public-demo.txt
Logging destination bucket	assignment-09-amzn-s3-logs-181026-5842
Log destination prefix	access-logs/
Final resource state	Both buckets emptied and deleted after evidence capture; source logging disabled before cleanup.

3. Logical Architecture

The source bucket held the demonstration objects and the configuration for versioning, lifecycle management, and public read access under one prefix. Server access logs were delivered to a different bucket in the same Region. The diagram is a logical summary of the configuration used during the exercise; the buckets were deleted after the tests.

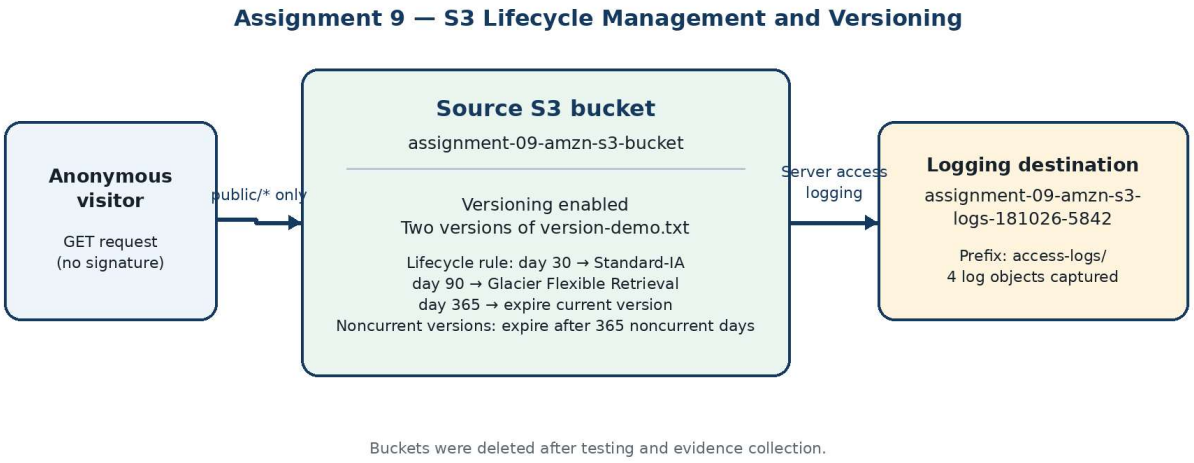


Figure 1. Logical architecture of the S3 lifecycle, access policy, and logging configuration.

4. Implementation and Evidence

4.1 Create the source bucket

A general-purpose S3 bucket named assignment-09-amzn-s3-bucket was created in Asia Pacific (Mumbai), ap-south-1. The initial object list was empty. Default private access settings were retained initially; public access was later granted only through a restricted bucket policy for public/*.

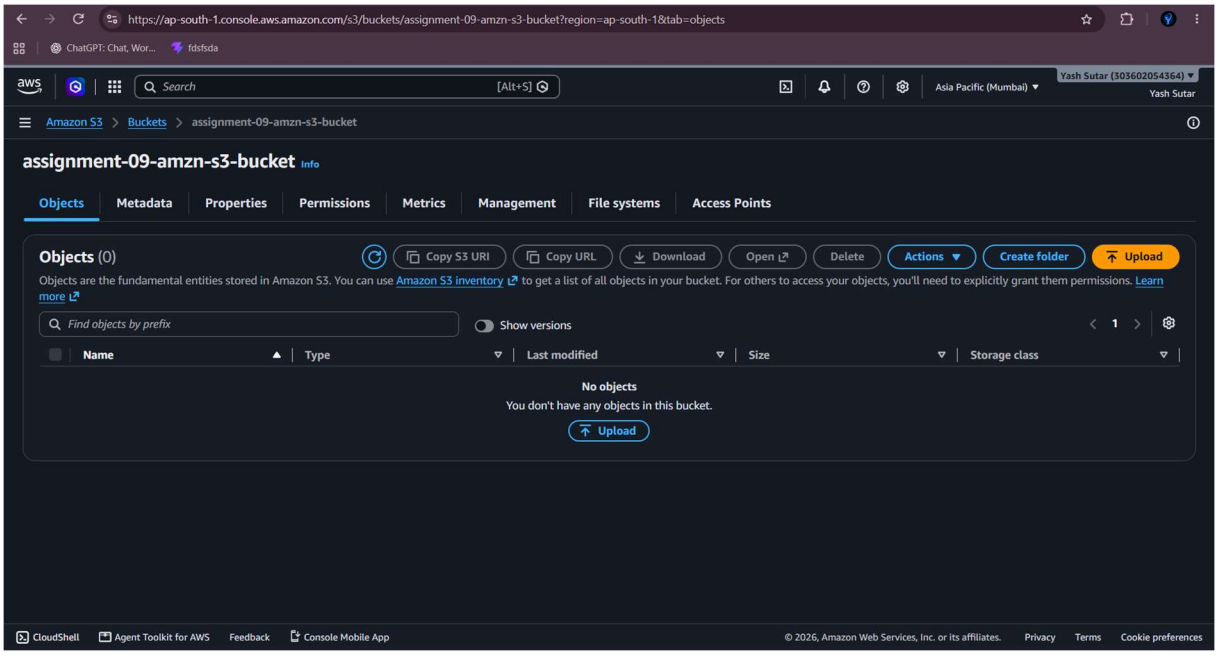


Figure 2. Source bucket created in the Mumbai Region.

4.2 Enable and demonstrate S3 Versioning

Bucket Versioning was enabled from the Properties tab. The file version-demo.txt was uploaded, its contents were changed from Content Version 1 to Content Version 2, and the changed file was uploaded again with the same object key. With Show versions enabled, S3 displayed two rows with different Version IDs, demonstrating that a new upload creates a new version rather than overwriting the prior version permanently.

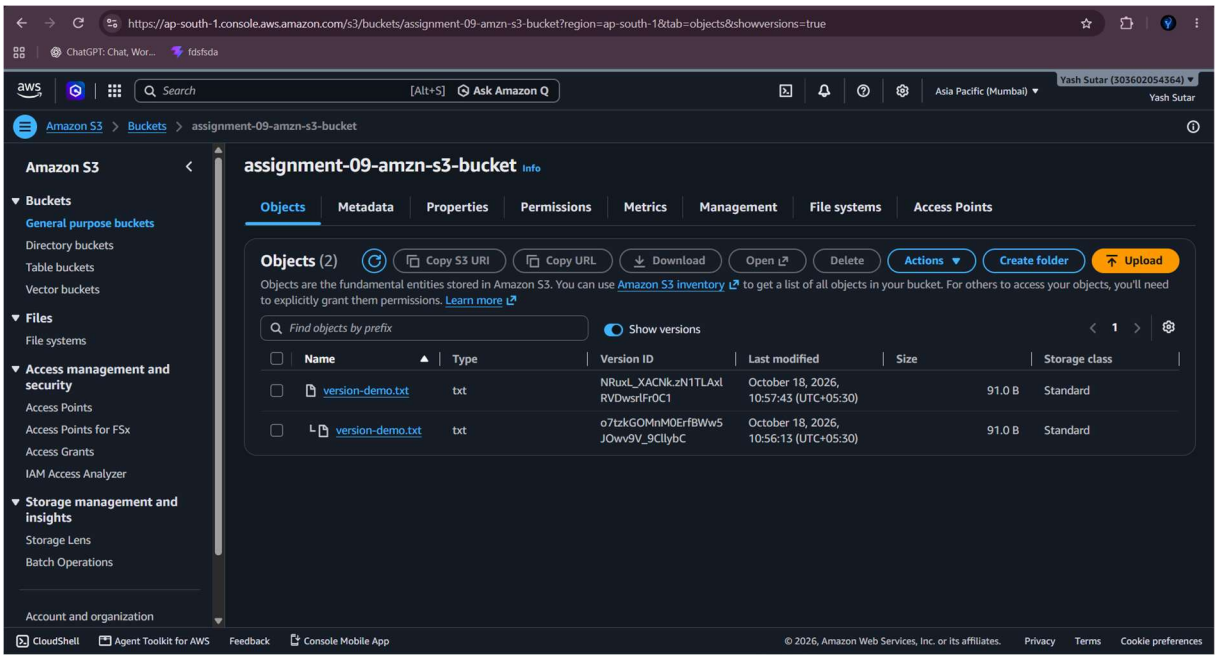


Figure 3. Two versions of version-demo.txt with distinct Version IDs.

Versioning note: an ordinary delete on a versioned bucket commonly adds a delete marker. The lifecycle configuration therefore includes NoncurrentVersionExpiration to manage older versions separately. Permanently expired noncurrent versions are not recoverable.

4.3 Configure lifecycle management

A lifecycle rule named a9-lifecycle-30-90-365 was created and left Enabled for the entire bucket. The lifecycle JSON submitted with this report records the selected transitions and expiration actions.

Lifecycle action	Configured rule value	Purpose
Transition current objects	Day 30 → STANDARD_IA (S3 Standard-IA)	Move eligible objects to infrequent-access storage.
Transition current objects	Day 90 → GLACIER (S3 Glacier Flexible Retrieval)	Move eligible objects to archival storage.
Expire current version	365 days after object creation	Make the current version expire; in a versioned bucket this normally adds a delete marker.
Expire noncurrent versions	365 days after becoming noncurrent	Permanently delete eligible older versions.

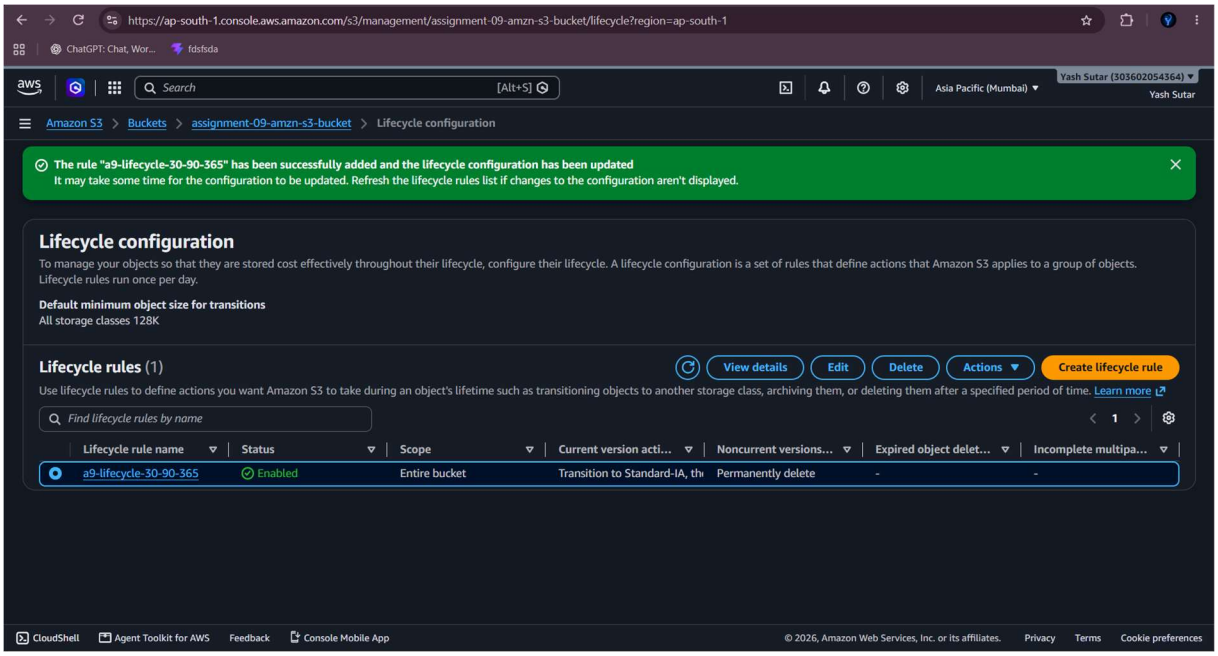


Figure 4. Lifecycle rule a9-lifecycle-30-90-365 saved and shown as Enabled for the entire bucket.

Operational note: the console showed a default minimum object size of 128 KB for lifecycle transitions. The sample files shown in the evidence were much smaller than 128 KB, so they were not suitable for demonstrating that a storage-class transition had executed. The configuration was saved and verified, but actual 30-day/90-day transitions and 365-day expiration could not be observed during this short lab.

4.4 Configure public read access for a specific prefix

A folder named public/ was created. The source bucket policy granted only s3:GetObject to Principal * for the resource arn:aws:s3:::assignment-09-amzn-s3-bucket/public/*. No public write, delete, or bucket-list permission was included. Block Public Access policy settings had to permit the narrowly scoped public bucket policy; ACL-based public access was not used.

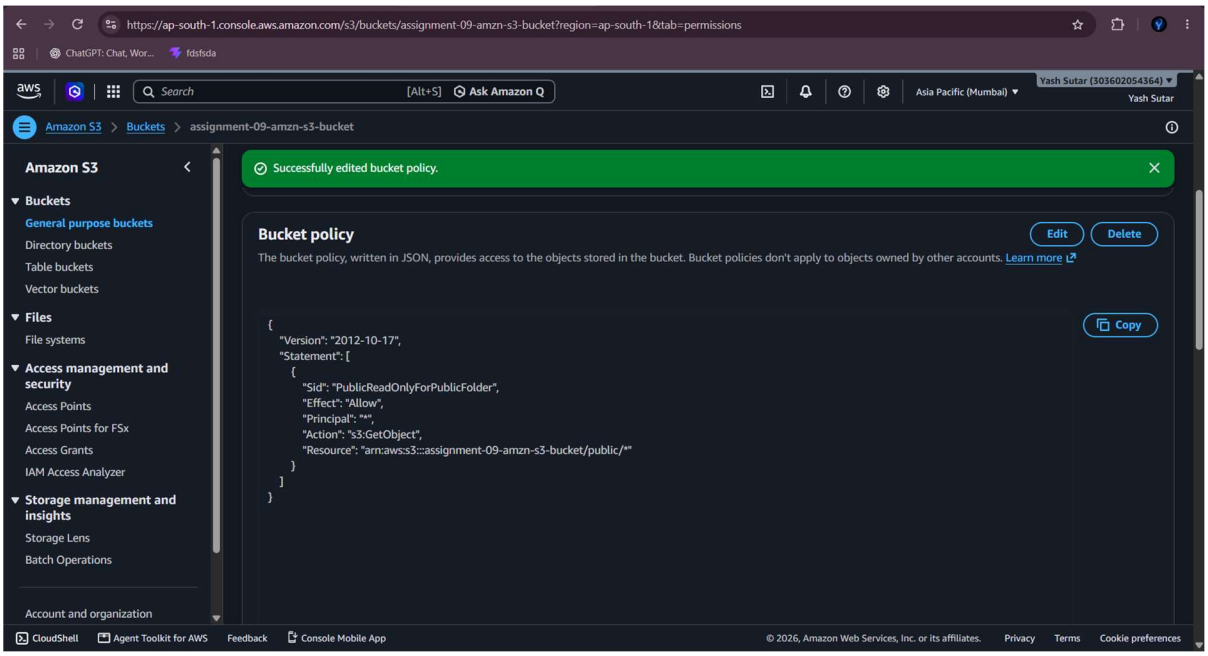


Figure 5. Saved bucket policy restricting public reads to the public/ prefix.

A harmless file named public-demo.txt was uploaded under public/.

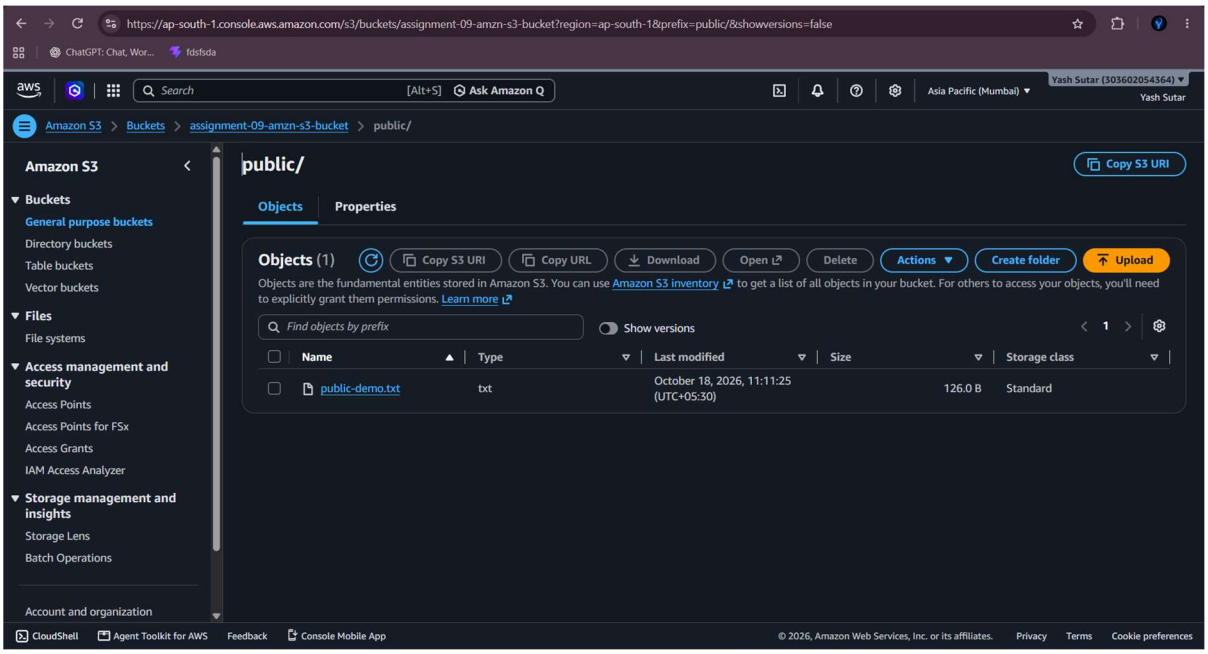


Figure 6. public-demo.txt uploaded inside the public/ prefix.

The clean unsigned object URL opened in an Incognito window and displayed the file contents. The user also reported completing the private-object test against version-demo.txt outside public/; it returned AccessDenied. A separate screenshot of that denial was not included in the evidence bundle.

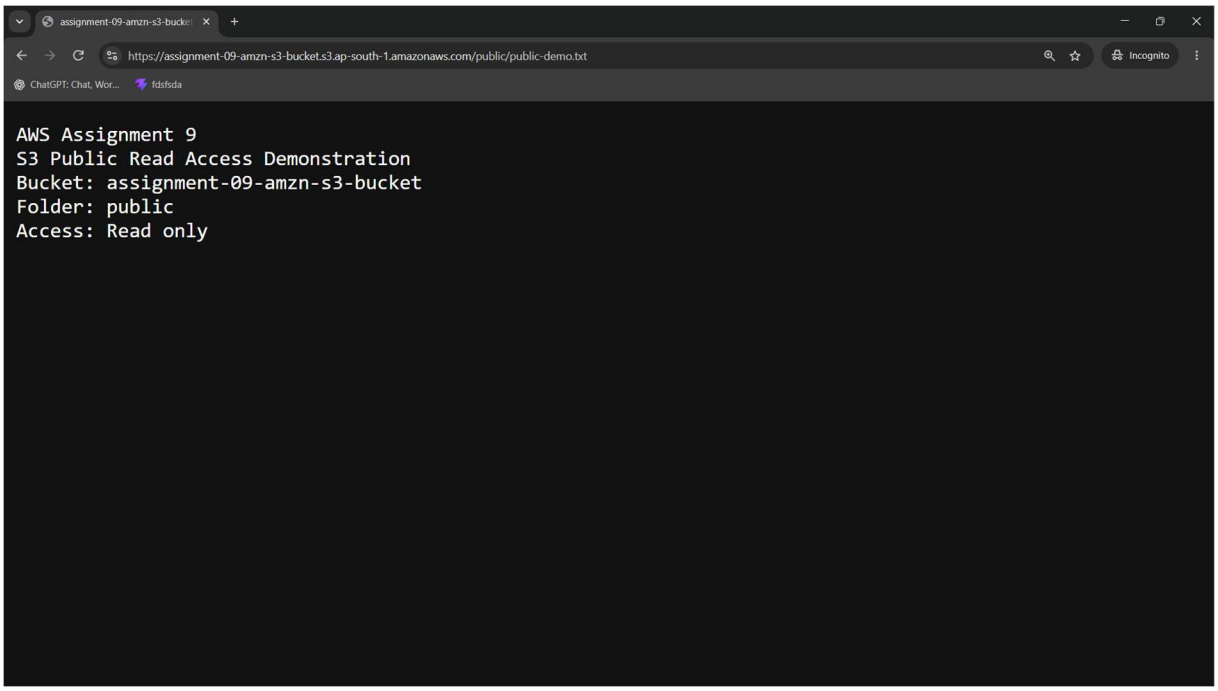


Figure 7. Anonymous browser access to public-demo.txt through the clean object URL.

4.5 Configure server access logging

A separate destination bucket, assignment-09-amzn-s3-logs-181026-5842, was created in the same Region. Its public access remained blocked. Server access logging was enabled on the source bucket with the destination set to the logging bucket and the prefix access-logs/. The console showed the log object key format beginning with access-logs/.

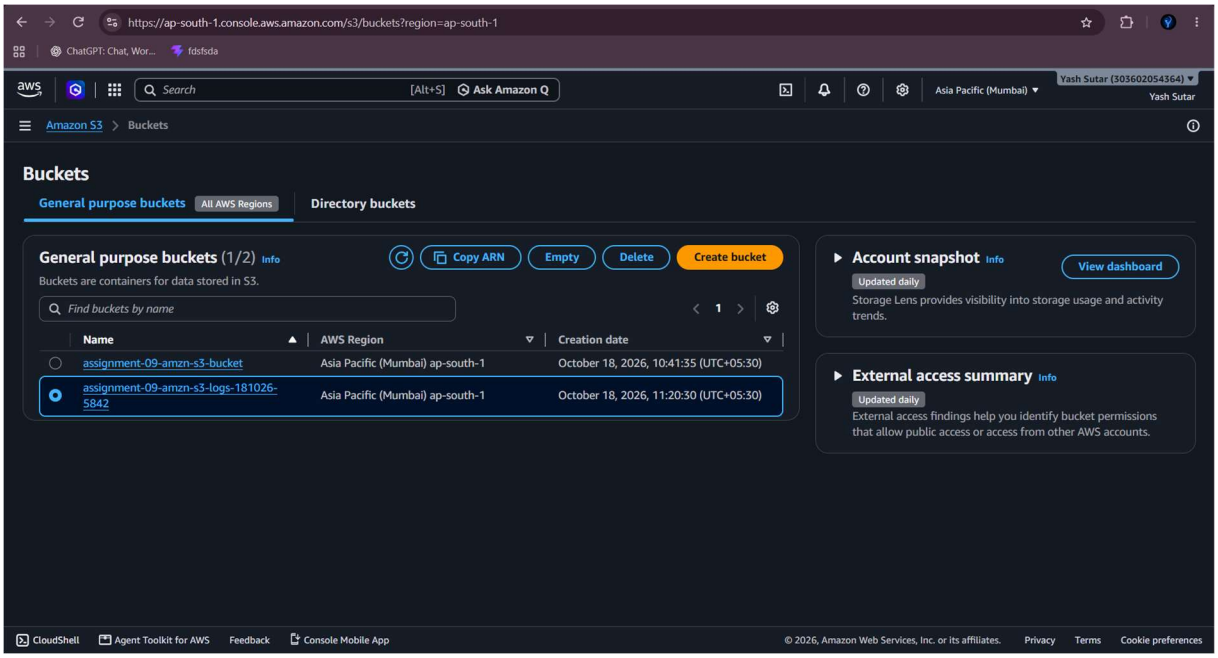


Figure 8. Source and separate logging destination buckets visible in the S3 bucket list.

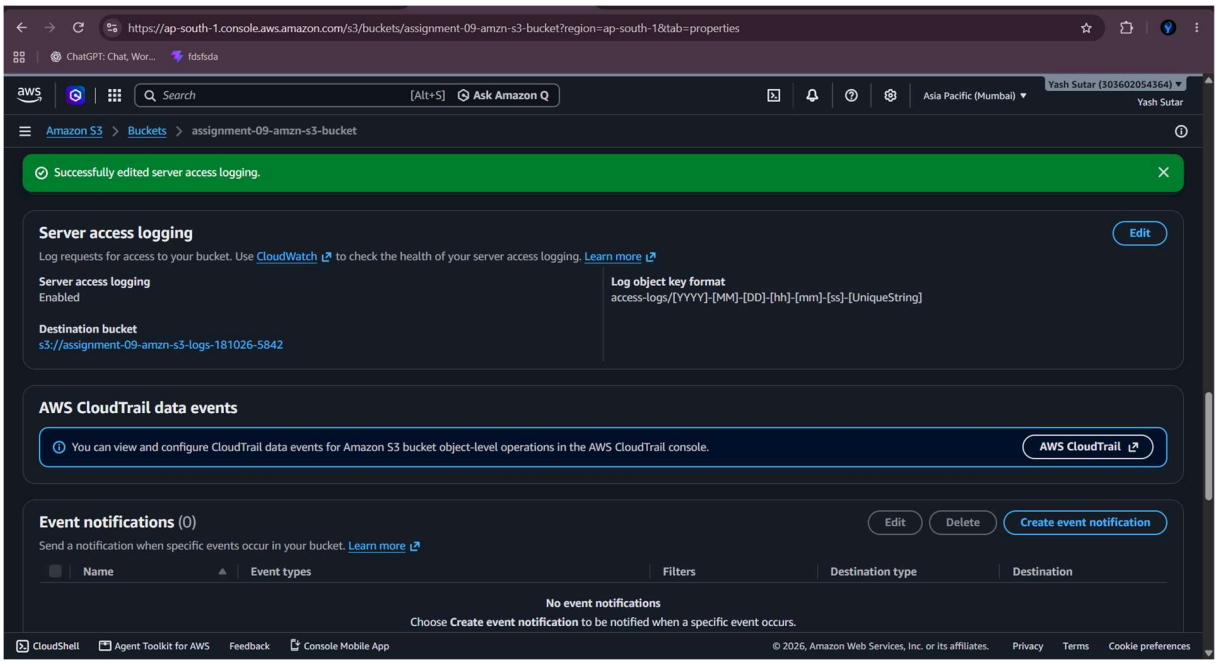


Figure 9. Server access logging successfully enabled with the destination bucket and access-logs/ prefix.

After generating requests against the source bucket and waiting for log delivery, four log objects appeared under the access-logs/ prefix. AWS server access logging is best-effort and delivery may take several hours; actual log arrival was observed in this run.

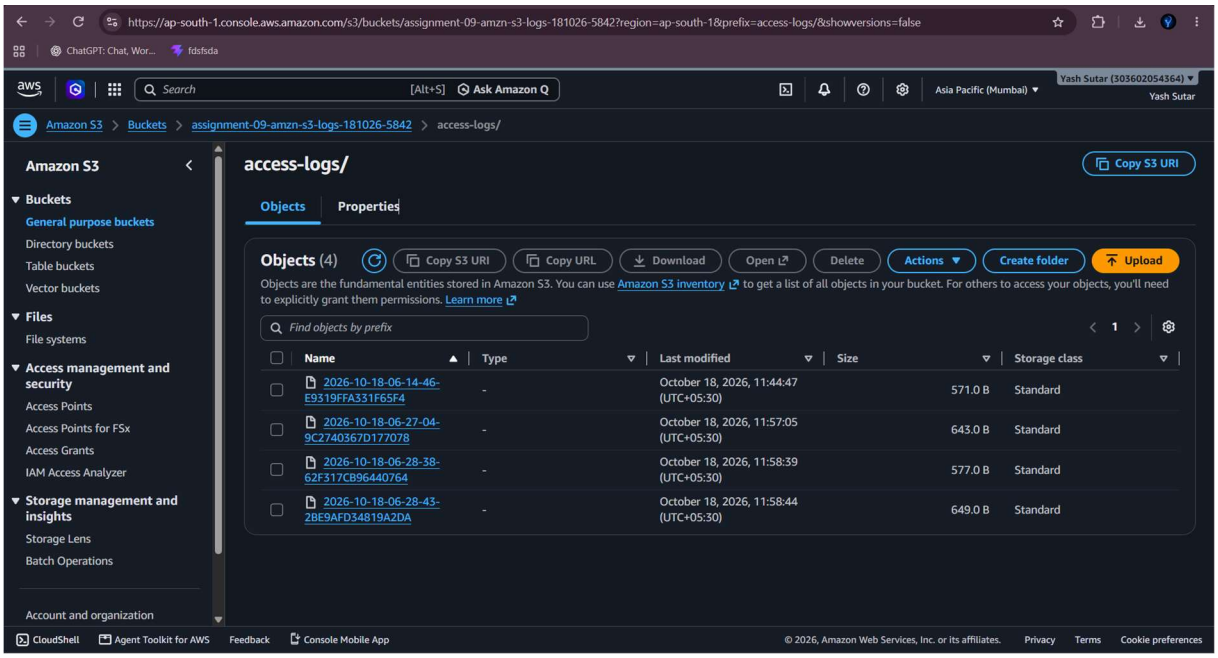


Figure 10. Four server access log objects delivered to the access-logs/ prefix.

5. Verification and Test Results

Test	Expected result	Observed result	Status
Bucket creation	Source bucket exists in ap-south-1	Bucket created; screenshot captured	PASS
Versioning	Repeated upload of same key retains versions	Two distinct Version IDs shown for version-demo.txt	PASS
Lifecycle rule	Rule enabled with 30/90/365-day actions and noncurrent cleanup	Rule a9-lifecycle-30-90-365 shown Enabled; JSON records the values	PASS — configuration
Public read	Object under public/ opens anonymously	public-demo.txt content shown in Incognito using clean URL	PASS
Private object read	Object outside public/ is not publicly readable	AccessDenied reported by user; no separate screenshot included	PASS — reported
Server access logging	Logs appear in separate destination under access-logs/	Four log objects were visible in the destination prefix	PASS
Cleanup	Stop logging and remove temporary AWS resources	Logging disabled; both buckets emptied and deleted	PASS

6. Challenges, Notes, and Limitations

- Lifecycle timing: lifecycle actions are age-based and asynchronous. A short practical session cannot wait 30, 90, or 365 days to observe those actions on live objects.
- Versioned expiration: expiration of a current version in a versioning-enabled bucket usually creates a delete marker. NoncurrentVersionExpiration manages permanent deletion of older versions after their configured noncurrent period.
- Log delivery: the destination was empty initially, but four log objects arrived after later requests and waiting. AWS describes server access logging as best-effort, so delivery timing and completeness are not guaranteed.
- Evidence after cleanup: the buckets were deleted after screenshots and policy files were saved. The configuration is therefore evidenced by the saved screenshots and JSON artifacts, not by a currently active AWS resource.

7. Cleanup and Cost Control

Following the assessment guidance to stop or delete AWS resources after testing, server access logging was disabled first. The source bucket was emptied and deleted. The logging destination bucket was then emptied (including the four delivered log objects) and deleted. These actions

removed the temporary S3 resources after the screenshots and JSON policy files had been collected.

The screenshot in Figure 10 documents that log delivery worked before cleanup. Because logs were intentionally removed from S3 during cleanup, the saved screenshot is the retained evidence of the delivered log objects.

8. Conclusion

The assignment requirements were implemented and verified: an S3 bucket was created, versioning was demonstrated with two object versions, a lifecycle rule was saved and enabled, public read access was limited to public/*, and server access logging delivered four log objects to a separate destination bucket. Both buckets were deleted after evidence collection to avoid leaving temporary resources active. The lifecycle JSON and bucket policy JSON are included in the appendices and as separate files.

Appendix A — Lifecycle Policy JSON

This is the lifecycle configuration saved as lifecycle-policy.json. It applies to all objects through an empty-prefix filter. The rule transitions current versions and manages both current expiration and noncurrent-version expiration.

```
{
  "Rules": [
    {
      "ID": "a9-lifecycle-30-90-365",
      "Status": "Enabled",
      "Filter": {
        "Prefix": ""
      },
      "Transitions": [
        {
          "Days": 30,
          "StorageClass": "STANDARD_IA"
        },
        {
          "Days": 90,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 365
      },
      "NoncurrentVersionExpiration": {
        "NoncurrentDays": 365
      }
    }
  ]
}
```

Separate deliverable file: lifecycle-policy.json

Appendix B — Public Folder Bucket Policy JSON

This bucket policy permits anonymous s3:GetObject requests for objects whose keys begin with public/. It does not grant listing, upload, or deletion permissions.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadOnlyForPublicFolder",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::assignment-09-amzn-s3-bucket/public/*"
    }
  ]
}
```

Separate deliverable file: bucket-policy-public-folder.json